

Notes: Jiang, Lustig, Van Nieuwerburgh & Xiaolan (JFE, 2026) Manufacturing Risk-Free Government Debt

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1 Big picture

- **Question.** What restrictions must government primary surpluses satisfy if government debt is truly risk-free, that is, if Treasury debt has zero beta with aggregate risk?
- **Main idea.** Risk-free government debt is not free. If bondholders are protected from aggregate shocks, taxpayers must absorb the risk through state-contingent taxes and primary surpluses.
- **Core trade-off.** A government can insure taxpayers over short horizons by issuing more debt in bad times. But if debt must stay risk-free and debt/output is mean-reverting, the same policy requires higher future surpluses in bad states.
- **Model object.** The paper combines a government debt valuation equation with a dynamic asset-pricing model and a calibrated process for the U.S. marketable debt/output ratio.
- **Empirical object.** Given observed U.S. debt/GDP dynamics, the model backs out the sequence of expected future primary surpluses required to keep debt risk-free.
- **Main rejection.** The risk-free debt restriction is strongly rejected in U.S. surplus data, especially after the Great Financial Crisis and the COVID period.
- **Quantitative headline.** As of end-2022, keeping U.S. debt risk-free requires cumulative primary surpluses of about 28% of GDP from 2023–2032, while CBO projections imply a cumulative deficit of about 24% of GDP.
- **Return evidence.** During the COVID shock, U.S. government debt delivered a real return of roughly –25% between March 2020 and October 2023. This is difficult to reconcile with genuinely risk-free debt.
- **Convenience yields.** Seigniorage/convenience revenues help at short horizons but do not eliminate the long-run trade-off: if debt remains outstanding and output risk is priced, someone must bear aggregate risk.
- **Takeaway.** Assuming risk-free government debt can overstate fiscal capacity. Safe debt is manufactured by shifting aggregate risk from bondholders to taxpayers; if taxpayers are protected instead, debt becomes risky.

2 Data and measurement

2.1 Core fiscal variables

The paper works with three government cash-flow objects:

- **Tax revenue claim:** a claim to future tax revenues T_t .
- **Government spending claim:** a claim to future government spending G_t .
- **Debt portfolio:** the market value of outstanding government debt D_t .

The primary surplus is

$$S_t = T_t - G_t. \quad (1)$$

The key scaled variables are the debt/output ratio, spending/output ratio, and surplus/output ratio:

$$d_t = \frac{D_t}{Y_t}, \quad x_t = \frac{G_t}{Y_t}, \quad s_t = \frac{S_t}{Y_t}. \quad (2)$$

Interpretation: The paper asks what s_t must look like if D_t is risk-free. Debt is the liability; the primary surplus is the cash flow backing it.

2.2 U.S. debt/output process

The benchmark calibration uses annual U.S. data from 1947 through 2023. The debt/output ratio is modeled as an AR(2) process in logs with countercyclical debt issuance:

$$\log d_t = \phi_0 + \phi_1 \log d_{t-1} + \phi_2 \log d_{t-2} - \lambda \varepsilon_t - \frac{1}{2} \lambda^2. \quad (3)$$

A negative output innovation has $\varepsilon_t < 0$, so $-\lambda \varepsilon_t > 0$ and debt/output rises in bad times.

Interpretation: This process captures two facts: U.S. debt/GDP is persistent and tends to rise when output is hit by adverse shocks.

2.3 Benchmark calibration

The key benchmark values are:

- maximum annual Sharpe ratio $\gamma = 1.29$;
- real risk-free rate $r = 1.62\%$;
- mean output growth $\mu = 3.00\%$;
- output growth volatility $\sigma = 2.35\%$;
- GDP risk premium in log $\gamma\sigma - \frac{1}{2}\sigma^2 = 3.00\%$;
- mean debt/output $\bar{d} = 37.92\%$;
- debt/output AR coefficients $\phi_1 = 1.38$ and $\phi_2 = -0.45$;
- sensitivity of debt/output to output shocks $\lambda = 2.53\sigma$;
- government spending/output AR coefficient $\varphi_1^g = 0.86$ and mean $\bar{x} = 17.56\%$.

Interpretation: The calibration is designed to match debt/GDP persistence, countercyclicity, and basic macro asset-pricing moments. The model is not fitted to observed future surpluses; those are used as a test.

2.4 Observed and projected surpluses

- The empirical surplus is the U.S. primary surplus as a share of GDP.
- The paper compares model-implied expected surpluses with realized surpluses and CBO projected primary surpluses.
- Since the GFC, U.S. realized and projected primary surpluses are far below those implied by the risk-free debt restriction.
- As of 2022, the model-implied 2023 surplus is about 8.1% of GDP, while the realized 2023 primary surplus is about -2.7% of GDP.

Interpretation: The empirical test is direct: feed observed debt/GDP into the risk-free debt model and ask whether the implied surplus path resembles the actual or CBO-projected path.

3 Models and empirical specifications

3.1 Government debt valuation

Under no arbitrage and a transversality/no-bubble condition, the market value of government debt equals the present discounted value of future primary surpluses:

$$D_t = P_t^T - P_t^G = \mathbb{E}_t \left[\sum_{j=1}^{\infty} M_{t,t+j} S_{t+j} \right], \quad (4)$$

where $M_{t,t+j}$ is the stochastic discount factor from t to $t+j$.

Interpretation: Debt is a claim backed by future net taxes. If the debt is risk-free, the required future surpluses are tightly constrained by the debt dynamics.

3.2 Return beta

For any asset i , the paper defines the return beta with respect to the pricing kernel as

$$\beta_t^i = - \frac{\text{cov}_t(M_{t+1}, R_{t+1}^i)}{\text{var}_t(M_{t+1})}. \quad (5)$$

The Euler equation implies

$$\mathbb{E}_t[R_{t+1}^i - R_t^f] = \beta_t^i \frac{\text{var}_t(M_{t+1})}{\mathbb{E}_t[M_{t+1}]}. \quad (6)$$

Interpretation: A zero-beta debt portfolio is risk-free in the asset-pricing sense: it does not lose value in bad aggregate states.

3.3 General fiscal trade-off

Let β_t^T , β_t^G , and β_t^D be the betas of the tax claim, spending claim, and debt portfolio. Proposition 1 states

$$\beta_t^T = \frac{P_t^G}{D_t + P_t^G} \beta_t^G + \frac{D_t}{D_t + P_t^G} \beta_t^D. \quad (7)$$

If debt is risk-free, $\beta_t^D = 0$, so

$$\beta_t^T = \frac{P_t^G}{D_t + P_t^G} \beta_t^G < \beta_t^G \quad \text{when } D_t > 0 \text{ and } \beta_t^G > 0. \quad (8)$$

Interpretation: To manufacture risk-free debt, the tax claim must be safer than the spending claim. In recessions, taxes must not fall too much, and may need to rise relative to GDP. Taxpayers insure bondholders.

3.4 Output and stochastic discount factor

Output follows permanent shocks:

$$y_{t+1} - y_t = \mu + \sigma \varepsilon_{t+1}, \quad \varepsilon_{t+1} \sim N(0, 1). \quad (9)$$

The log stochastic discount factor is

$$m_{t,t+1} = -\rho - \frac{1}{2} \gamma^2 - \gamma \varepsilon_{t+1}. \quad (10)$$

Interpretation: Bad output shocks are also high marginal-utility states. The market prices permanent output risk, so shifting risk between taxpayers and bondholders is economically important.

3.5 Constant debt/output benchmark

With constant spending/output $x = G_t/Y_t$ and constant debt/output $d = D_t/Y_t$, the government budget constraint implies a primary surplus/output ratio

$$s_t = -d[1 - \exp\{-(\mu - \rho + \sigma\varepsilon_t)\}]. \quad (11)$$

Interpretation: If debt is risk-free, a bad output shock requires higher taxes or lower net transfers so that the debt payoff remains protected. The tax burden must move countercyclically.

3.6 Countercyclical debt issuance

The richer model lets debt/output move persistently and countercyclically:

$$\log d_t = \phi_0 + \sum_{p=1}^P \phi_p \log d_{t-p} - \lambda\varepsilon_t - \frac{1}{2}\lambda^2. \quad (12)$$

For the benchmark $P = 2$, $\phi_1 = 1.38$ and $\phi_2 = -0.45$.

Interpretation: Countercyclical debt issuance lets the government run deficits immediately after bad shocks. But mean reversion in debt/output means those deficits must eventually be reversed by surpluses.

3.7 Expected future surpluses under risk-free debt

For $P = 2$, Proposition 5 gives

$$\mathbb{E}_t[s_{t+j}] = \mathbb{E}_t[d_{t+j-1}] \left[\exp\left(\rho - \mu + \frac{\sigma^2}{2}\right) - \exp(\log \mathbb{E}_t[d_{t+j}] - \log \mathbb{E}_t[d_{t+j-1}]) \right]. \quad (13)$$

The sensitivity to current debt is

$$\frac{\partial \mathbb{E}_t[s_{t+j}]}{\partial (\log d_t - \bar{d})} = \psi_{1,j-1} \mathbb{E}_t[s_{t+j}] + (\psi_{1,j-1} - \psi_{1,j}) \mathbb{E}_t[d_{t+j}]. \quad (14)$$

Interpretation: The expected surplus path depends on the risk-free rate, expected output growth, and debt/output dynamics. It does not depend on the market price of risk except through the risk-free rate.

3.8 Impulse response of surpluses

For the benchmark $P = 2$ case and $\rho = \mu$, Proposition 6 states that the impulse response of the surplus/output ratio to an output shock is

$$\frac{\partial s_{t+j}}{\partial \varepsilon_{t+1}} = (\lambda - \sigma)e^{\bar{d}}, \quad j = 1, \quad (15)$$

$$= \lambda(\phi_1 - 1)e^{\bar{d}}, \quad j = 2, \quad (16)$$

$$= \lambda(\psi_{1,j-1} - \psi_{1,j-2})e^{\bar{d}}, \quad j \geq 3. \quad (17)$$

Interpretation: The model permits short-run taxpayer insurance if debt rises after bad shocks. But the sign eventually flips: surpluses must rise later to restore debt/output and preserve risk-free debt.

3.9 Cash-flow betas over finite horizons

For a generic h -period stream of discounted cash flows Z , the cash-flow beta is

$$\beta_t^{Z,CF}(h) = -\frac{\text{cov}_t(M_{t+1}, \mathbb{E}_{t+1}[\sum_{j=1}^h M_{t+1,t+j} Z_{t+j}])}{D_t \text{var}_t(M_{t+1})}. \quad (18)$$

For surpluses, Proposition 7 gives

$$\beta_t^{S,CF}(h) = \frac{\text{cov}_t(M_{t+1}, \mathbb{E}_{t+1}[M_{t+1,t+h} D_{t+h}])}{D_t \text{var}_t(M_{t+1})}. \quad (19)$$

Interpretation: When current debt is risk-free, the risk properties of cumulative future surpluses over horizon h are pinned down by the riskiness of future debt at horizon h .

3.10 Government spending and convenience yields

The spending/output ratio follows

$$\log x_t = \varphi_0^g + \varphi_1^g \log x_{t-1} - b_g \varepsilon_t - \frac{1}{2} b_g^2. \quad (20)$$

The model also considers convenience yields. If safe debt produces a seigniorage/convenience revenue stream, the surplus cash-flow beta is adjusted by the cash-flow beta of seigniorage revenues:

$$\beta_t^{S,CF}(h) = \text{debt-strip beta component} - \beta_t^{K,CF}(h). \quad (21)$$

Interpretation: Convenience yields can relax the constraint at short horizons, but in the long run seigniorage revenues are themselves tied to debt/output and output, so they do not eliminate taxpayer risk.

4 Identification and empirical design

4.1 What is being tested

The paper tests the joint restrictions implied by three assumptions:

- government debt satisfies the no-bubble/transversality valuation condition;
- the government debt portfolio is risk-free, i.e., $\beta_t^D = 0$;
- observed U.S. debt/output dynamics are well represented by the calibrated AR(2) process.

Interpretation: If these assumptions are true, the model pins down expected future primary surpluses. The empirical question is whether realized and projected surpluses look like those implied values.

4.2 Why the test is sharp

- The expected-surplus restriction does not require estimating all pricing-kernel parameters.
- Given d_t , r , g , and the debt/output dynamics, the model directly computes $\mathbb{E}_t[s_{t+j}]$.
- The model's implications are especially strong after debt/output rises far above its long-run mean.
- The post-GFC and COVID periods therefore provide powerful tests.

Interpretation: The model is not merely saying that high debt eventually requires some adjustment. It provides a quantitative path and timing for the adjustment required to make debt risk-free.

4.3 Four empirical contradictions

1. **Level of surpluses:** risk-free debt implies large expected surpluses after high debt episodes; realized surpluses are far lower.
2. **CBO projections:** projected primary surpluses also fall far below what the model requires.
3. **Impulse-response timing:** the model allows only limited short-run deficits after adverse shocks; U.S. deficits remain large and persistent after the GFC and COVID.
4. **Debt returns:** U.S. government debt suffered large real losses in the COVID period, which suggests bondholders were not fully insulated from aggregate risk.

Interpretation: The paper’s rejection is cumulative: expected surplus levels, CBO forecasts, dynamic responses, and realized debt returns all point against the risk-free debt model.

4.4 Robustness and extensions

- The results generalize from AR(2) debt dynamics to higher-order AR(P) dynamics.
- Allowing convenience yields weakens the constraint over short horizons but not at long horizons.
- Richer asset-pricing models with disaster risk deliver similar qualitative implications.
- Models with transitory output risk still impose strong restrictions because long-term debt carries interest-rate risk.
- Appendix tests using simulation-based standard errors and time-varying growth/rate expectations continue to reject the risk-free debt model.

Interpretation: The paper’s main message is not an artifact of a single calibration. The trade-off between taxpayer insurance and bondholder insurance is structural.

5 Tables and figures

5.1 Table 1 and Figure 1: benchmark calibration and debt/output autocorrelation

Read: Table 1:

- The real risk-free rate is calibrated at 1.62%, mean output growth at 3.00%, and output growth volatility at 2.35%.
- The debt/output ratio has mean 37.92%, high persistence, and AR(2) coefficients $\phi_1 = 1.38$ and $\phi_2 = -0.45$.
- Debt/output is countercyclical: a one-standard-deviation output shock is associated with debt/output sensitivity $\lambda = 2.53\sigma$.

Read: Figure 1:

- The AR(2) calibration closely matches the empirical autocorrelation and partial autocorrelation of log debt/GDP.
- The ACF decays slowly, showing high persistence.

- The PACF has a large first lag and negative second lag, consistent with the AR(2) specification.

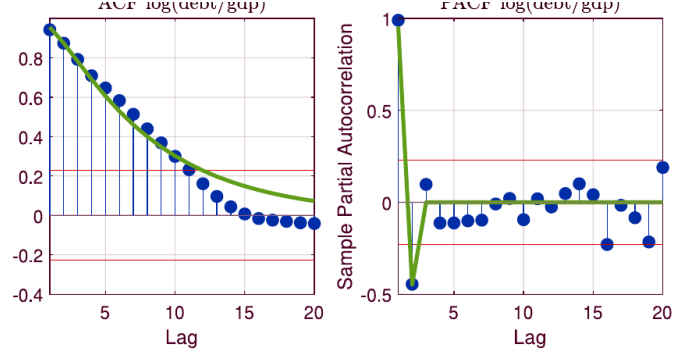
Economic message: The calibration gives the model its discipline: debt/GDP is persistent, stationary, and countercyclical. Risk-free debt then imposes strong surplus restrictions.

Table 1

Benchmark calibration for U.S.

Panel A: Preferences and Output Dynamics		
γ	1.29	maximum annual Sharpe ratio
r	1.62%	real risk-free rate
μ	3.00%	mean of growth rate of output
σ	2.35%	std. of growth rate of output
$rp = \gamma\sigma - \frac{1}{2}\sigma^2$	3.00%	GDP risk premium in log
Panel B: Debt/Output Ratio Dynamics		
λ	$2.53 \times \sigma$	sensitivity of debt/output to output innovations
$d = \exp\{\phi_0/(1 - \phi_1 - \phi_2)\}$	37.92%	mean of debt/output
ϕ_1	1.38	first-order coeff of debt/output
ϕ_2	-0.45	second-order coeff of debt/output
Panel C: Government Spending/Output Ratio Dynamics		
b^s	$1.42 \times \sigma$	sensitivity of spending/output to output innovations
φ_1^s	0.86	AR(1) coeff of spending/output
$x = \exp\{\varphi_0^s/(1 - \varphi_1^s)\}$	17.56%	mean of govt. spending/output

(a) Table 1: benchmark calibration.



tion in Debt/Output.
sample autocorrelation and partial autocorrelation of the U.S. log government debt/output ratio. The sample is annual from 194 : data are denoted by blue dots and the model-implied moments are denoted by the solid green line. The parameters are listed

(b) Figure 1: autocorrelation in debt/output.

5.2 Figure 2 and Figure 3: surplus dynamics and model-implied surpluses

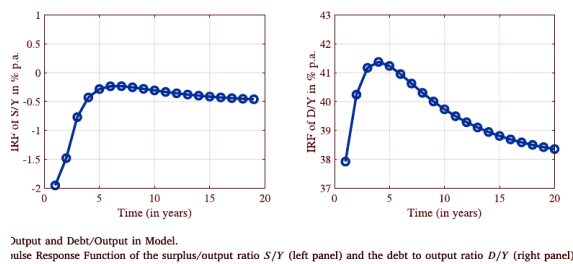
Read: Figure 2:

- After a negative output shock, the model allows short-run deficits and a hump-shaped rise in debt/output.
- The surplus response turns positive after a few years because debt must revert to its long-run mean.
- The government can delay fiscal adjustment but cannot avoid it while preserving risk-free debt.

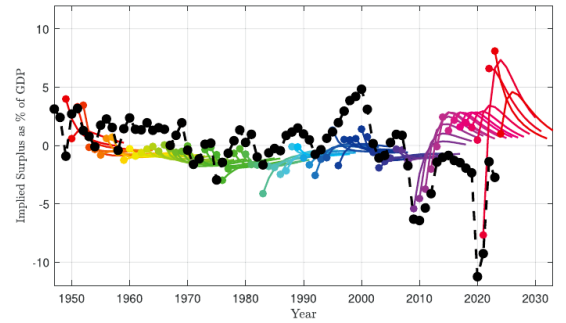
Read: Figure 3:

- Model-implied expected surpluses rise strongly after the GFC and COVID debt expansions.
- Realized surpluses and CBO projections are far below the required path.
- The 2022 calibration implies about 8.1% of GDP surplus in 2023 and cumulative 2023–2032 surpluses of about 28% of GDP.

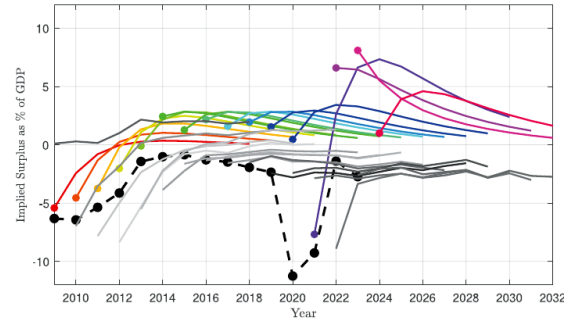
Economic message: The risk-free debt model says high debt must be followed by large primary surpluses. The U.S. fiscal path after the GFC and COVID does not satisfy that restriction.



(a) Figure 2: IRF of surplus/output and debt/output.



(a) Full Time-Series



(b) Since the GFC

(b) Figure 3: expected surpluses.

5.3 Figure 4 and Figure 5: risk premia across horizons and the role of saving

Read: Figure 4:

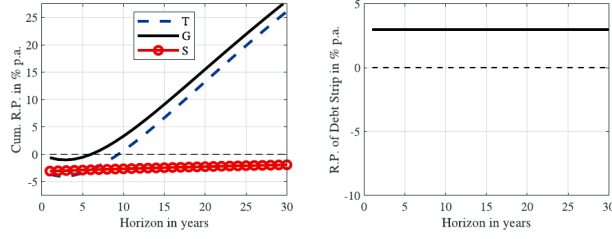
- Panel A, with constant debt/output, shows no meaningful taxpayer insurance: surplus risk premia are negative and taxpayers insure bondholders.
- Panel B, with AR(2) debt/output, shows short-run taxpayer insurance but only for a limited period.
- The surplus risk premium eventually turns negative, reflecting the future tax burden needed to protect bondholders.

Read: Figure 5:

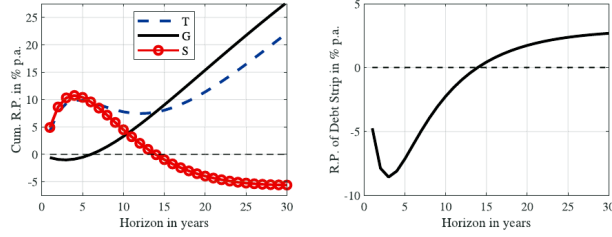
- If the government saves rather than borrows, it can generate a different risk-sharing pattern.
- The figure shows that saving can protect both taxpayers and bondholders more effectively than positive debt.
- The key logic is that positive outstanding debt makes taxpayers the residual risk bearers when debt is risk-free.

Economic message: Countercyclical debt issuance gives short-run relief but creates long-run risk for taxpayers. Saving is the clean way to relax the constraint.

Panel A: Constant Debt/Output ($\lambda = 0$)



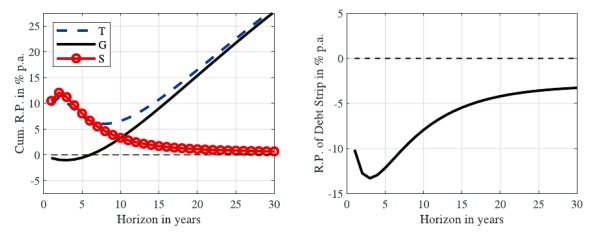
Panel B: AR(2) Debt/Output ($\phi_1 = 1.38, \phi_2 = -0.45$)



remia across horizons.

(a) Figure 4: risk premia across horizons.

AR(2) Savings/Output



remia across horizons with saving.

ts the risk premium of cumulative discounted cash flows, $\beta^t CF_t(h) \times \frac{1}{1+r}$, in the left panel against the horizon h . The right panel

(b) Figure 5: risk premia with saving.

5.4 Figure 6 and Table B.1: convenience yields and statistical rejection

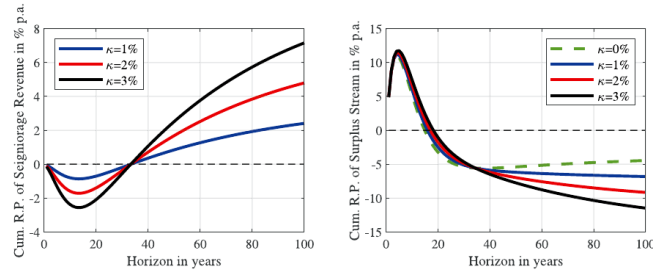
Read: Figure 6:

- Seigniorage revenues from convenience yields can reduce short-run taxpayer burden.
- Higher convenience yields create a larger short-run effect.
- At long horizons, seigniorage risk becomes tied to debt/output and output, so the long-run trade-off remains.

Read: Table B.1:

- Regressions show realized surpluses are statistically different from model-implied surpluses.
- In the full sample, realized surplus averages are below implied surplus averages.
- In the post-2008 sample, the rejection is stronger and CBO projections also fall below the model-implied path.

Economic message: Convenience yields are helpful but not a panacea. The appendix regressions quantify the gap between risk-free-debt-implied surpluses and actual/projected surpluses.



premia on seigniorage and surplus claims in convenience yields model.

(a) Figure 6: convenience-yield model.

Table B.1

Regression of implied surpluses on realized surpluses.

Panel A: Full Sample, Realized Surplus					
Data Surplus	Model Surplus	Intercept	p-value	Slope	p-value
-0.27	-0.51	-0.43	0.00	-0.30	0.00
Panel B: Post-2008 Sample, Realized Surplus					
Data Surplus	Model Surplus	Intercept	p-value	Slope	p-value
-3.66	1.48	-4.14	0.00	0.33	0.05
Panel C: Post-2008 Sample, CBO Surplus Projections					
CBO Surplus	Model Surplus	Intercept	p-value	Slope	p-value
-1.15	1.73	-0.89	0.00	-0.15	0.01

Notes: Panel A and B: Regression of the realized surplus/output ratios in year $t+j$ on the model's implied expected surplus/output ratios for that year, based on the information available in year t . The regression equation is given by: $s_{t+j} = \alpha + \beta \mathbb{E}_t[s_{t+j}] + \epsilon_{t+j}$. The sample includes all horizons j from 1 to 10 and all years t from 1947 until 2022 (pooled regression). The p -value in column (4) tests the null that realized and predicted

(b) Table B.1: regressions of implied on realized surpluses.

5.5 Figures B.1 and B.2: simulation and time-varying expectations

Read: Figure B.1:

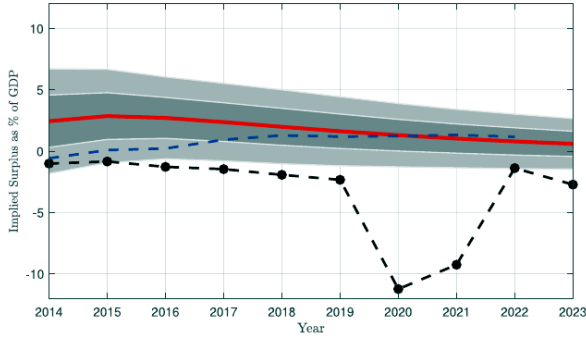
- Simulation-based confidence intervals around the risk-free model's surplus predictions do not contain realized surpluses in key post-GFC and post-COVID episodes.
- Using information as of 2013, actual surpluses fall below the model-implied distribution.
- Using information as of 2022, the CBO path and realized 2023 surplus are again below the model-implied path.

Read: Figure B.2:

- Allowing time-varying growth and rate expectations does not close the gap.
- Model-implied surpluses remain systematically above realized surpluses after the GFC.

Economic message: The rejection is not driven by ignoring uncertainty around the model prediction or by fixing growth and rate expectations.

Panel A: Using Information as of 2013



Panel B: Using Information as of 2022

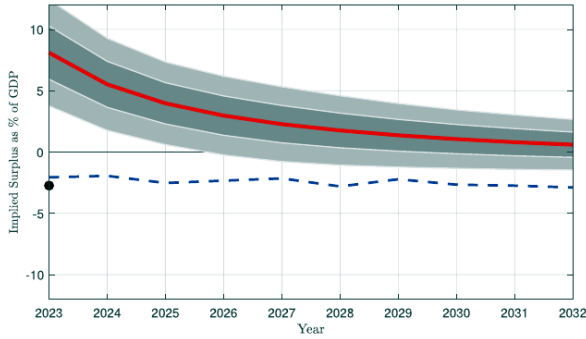


Fig. B.1. Model's implied surpluses and realized surpluses.

(a) Figure B.1: model-implied vs. realized surpluses.

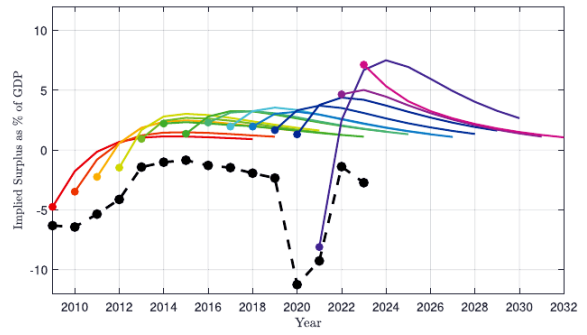


Fig. B.2. Model's implied surpluses and realized surpluses, with time-varying growth and rate expectations.

(b) Figure B.2: time-varying growth/rate expectations.

6 Short summary

- The paper studies the fiscal restrictions implied by risk-free government debt.
- Government debt is backed by future primary surpluses under no arbitrage and a transversality condition.
- If debt is risk-free, bondholders are insulated from aggregate shocks.
- Insulating bondholders forces taxpayers to bear aggregate risk through state-contingent taxes and surpluses.
- In a general trade-off, the tax claim beta is a weighted average of the spending-claim beta and debt beta.
- When debt beta is zero, the tax claim must be safer than the spending claim.
- The model uses permanent output shocks and a calibrated SDF with a realistic price of output risk.
- U.S. debt/output is modeled as a persistent, stationary, countercyclical AR(2) process.
- Countercyclical debt issuance lets the government run deficits briefly after bad shocks.
- But mean reversion in debt/output requires future surpluses to preserve risk-free debt.
- The model-implied expected surplus path is far above realized U.S. surpluses after the GFC and COVID.
- CBO projections are also far below the path required to keep debt risk-free.
- Appendix tests and robustness exercises confirm the rejection.
- Convenience yields ease the short-run constraint but do not eliminate the long-run taxpayer-bondholder trade-off.
- The central implication is that treating government debt as risk-free can overstate fiscal capacity.

7 Conclusion

7.1 Core conclusion

Jiang, Lustig, Van Nieuwerburgh, and Xiaolan show that risk-free government debt imposes strong restrictions on future primary surpluses. A government can make debt safe only by ensuring that taxpayers absorb enough aggregate risk. In U.S. data, especially after the GFC and COVID, the required surplus path is far above observed and projected surpluses.

7.2 Economic intuition

Risk-free debt protects bondholders in bad states. But aggregate risk does not disappear. If spending is countercyclical and debt is issued in recessions to protect taxpayers, future taxes or lower transfers must eventually rise enough to protect bondholders. Short-run insurance for taxpayers is possible, but long-run insurance for both taxpayers and bondholders is not possible when debt remains positive and output risk is priced.

7.3 Takeaway

The paper reframes fiscal sustainability. The issue is not only whether $r < g$ or whether debt/GDP is stationary. The deeper question is who bears aggregate risk. If policymakers assume Treasuries are risk-free, they implicitly assume taxpayers are the residual insurers. If taxpayers are protected instead, government debt becomes risky.